

Dean's conjecture at five

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Abstract

We prove that every nonempty finite simple graph of minimum degree at least five contains a cycle whose length is divisible by five. The proof is by contradiction. The two-connected case is ruled out by a theorem of Bai, Grzesik, Li, and Prorok, so we pass to a nontrivial end block B with cut vertex c and set $J = B - c$. At most four vertices of J have lost one unit of degree, and all are adjacent to c . After proving that J is 2-connected and then 3-connected, we repeatedly reintroduce c in rooted auxiliary graphs to repair precisely these degree deficits. This *root lifting* forces 4-connectivity and excludes K_4^- , triangles, and four-cycles. Thus J has girth at least six. A minimum theta, together with its exterior, then provides two three-path families whose residues yield the final contradiction modulo five. Together with the previously known cases, this establishes Dean's conjecture for every $k \geq 3$.

Formal verification. The modulus-five theorem is verified in Lean. The paper and Lean sources are available at github.com/pkchv/dean-conjecture. Section 1.3 explains the relationship between the proof presented here, the published results it uses, and the formal proof.

AI disclosure. The mathematical proof, manuscript, and Lean formalization were generated entirely by GPT-5.6 Sol and GPT-5.6 Sol Pro.

1. Introduction, proof architecture, and published results

Dean conjectured that, for every integer $k \geq 3$, every nonempty finite simple graph of minimum degree at least k contains a cycle whose length is divisible by k .^[5] The cases $k = 3$ and $k = 4$ were proved by Chen and Saito^[2] and Dean, Lesniak, and Saito^[6], respectively. Luo, Ma, and Zhao recently resolved every case $k \geq 6$.^[8] Thus $k = 5$ was the unique remaining case, as also recorded by Choi, Chu, Kim, and Park.^[4] The theorem proved here therefore completes Dean's conjecture.

For brevity, write BGLP for Bai–Grzesik–Li–Prorok, COY for Chiba–Ota–Yamashita, and GHLM for Gao–Huo–Liu–Ma.

1.1. Proof architecture

The proof is by contradiction. BGLP Theorem 1.3, together with direct cycles in its two exceptional families, disposes of a two-connected counterexample. We therefore work in a nontrivial end block B with cut vertex c . Every vertex of $B - c$ has degree at least five in B , while $2 \leq d_B(c) \leq 4$. Hence, for $J = B - c$, every vertex has degree at least four, and the degree-four vertices form a set

$$D \subseteq N_B(c), \quad |D| \leq 4.$$

In the auxiliary graphs constructed below, after the prescribed boundary losses are accounted for, precisely these vertices can remain one degree short of a rooted admissible-path theorem.

Under these standing hypotheses, the structural reduction follows the chain

$$J \text{ 2-connected} \implies J \text{ 3-connected} \implies J \text{ 4-connected} \implies \text{girth}(J) \geq 6.$$

The first implication uses BGLP's connectivity lemma. For the second and for the local exclusions that follow, the recurring device is root lifting: when there is no deficient vertex we apply GHLM directly, and when there is one we use it as the COY exception. When there are at least two, we adjoin c and restore one edge at each; the COY theorem then permits c as its single exceptional vertex. This yields admissible path families on opposite sides of a separator or outside a small core; joining those families produces five admissible cycles and hence a cycle of length 0 modulo five.

Once triangles and four-cycles are excluded, the standing counterexample assumption also excludes 5- and 10-cycles, so J has girth at least six and no 10-cycle. Choose a minimum-order theta H . GHLM Lemma 5.10 places H in a small induced envelope T ; the elementary subdivision-attachment lemma proved in Section 7 then shows that every vertex outside T has at most one neighbor in T . We prove that the exterior $M = J - V(T)$ is 2-connected. Lemma 2.4 supplies three x, y -paths in T with distinct residues modulo five, while root lifting supplies three admissible paths through M , possibly using c . The resulting nine cycles contradict the $3 + 3$ residue lemma, Lemma 2.3.

1.2. Notation

All graphs are finite, undirected, and simple; any graph subject to a minimum-degree hypothesis is nonempty. Paths are simple unless explicitly called walks, and length means number of edges. For a graph G , write $d_G(v)$ and $N_G(v)$ for the degree and neighborhood of v , and write $\delta(G)$ and $\kappa(G)$ for the minimum degree and vertex connectivity. If $X \subseteq V(G)$, put

$$d_X(v) = |N_G(v) \cap X|, \quad N_X(Q) = N_G(Q) \cap X$$

when the ambient graph is clear, where $N_G(Q) = \left(\bigcup_{v \in Q} N_G(v)\right) \setminus Q$. Thus $d_H(v)$ denotes degree in the graph H , whereas $d_X(v)$ counts ambient neighbors in the vertex set X .

The graph $G - X$ is obtained by deleting X , and $G[X]$ is induced by X . The notation $G - xy$ deletes the edge xy if present and otherwise leaves G unchanged; similarly, $G + xy$ adds the edge if absent and otherwise leaves G unchanged. A k -vertex cut has exactly k distinct vertices. Two edges are **independent** if they have no common endpoint. We use the standard block convention that a 2-connected graph is its unique end block and that all its vertices are inner vertices.

A **theta** is the union of three distinct, internally vertex-disjoint paths with the same two distinct ends. A sequence of paths or cycles is **admissible** if its lengths form an arithmetic progression with common difference one or two; for paths, the shortest length is at least two.

We use without further comment that a 2-connected graph which is not a cycle contains a theta, that an induced theta is 2-connected and has maximum degree at most three, and that adjoining a vertex with two distinct neighbors to a 2-connected graph preserves 2-connectivity. The required forms of the block-cut-tree argument, edge contraction, and Menger's theorem appear where they are used.

1.3. Published results used in the modulus-five proof

The modulus-five proof uses the following five published results.

Rooted admissible-path theorem (GHLM, Theorem 3.1).^[7] Let q be a positive integer, and let A have distinct roots x, y such that $A + xy$ is 2-connected. If

$$d_A(v) \geq q + 1 \quad (v \in V(A) \setminus \{x, y\}),$$

then A contains q admissible x, y -paths.

One-exception rooted theorem (the specialization of COY, Theorem 3, stated as BGLP, Theorem 2.2).^{[3], [1]} Let q be a positive integer, let $|V(A)| \geq 4$, and let $x, y, z \in V(A)$, with $x \neq y$, such that $A + xy$ is 2-connected. If

$$d_A(v) \geq q + 1 \quad (v \in V(A) \setminus \{x, y, z\}),$$

then A contains q admissible x, y -paths. No distinctness condition is imposed on z relative to the roots.

Connectivity lemma (BGLP, Lemma 2.3).^[1] Let $k \geq 2$ and $r \geq 0$ be integers. If a 2-connected graph H has minimum degree at least $k - r$ and has no k admissible cycles, then

$$\kappa(H) \geq \frac{k + 2}{2} - r.$$

Two-connected minimum-degree theorem (BGLP, Theorem 1.3).^[1] Let H have order n . If H is 2-connected and $\delta(H) \geq k \geq 4$, then H has k admissible cycles, unless it is isomorphic to one of the following two graphs:

$$H \cong K_{k+1} \quad \text{or} \quad H \cong K_{k, n-k}.$$

Minimum-theta structure (GHLM, Lemma 5.10).^[7] Let A have girth at least six and no 10-cycle, and let H be a minimum-order theta in A . Then H is induced; at most one vertex of $A - V(H)$ has at least two neighbors in H ; and, if such a vertex v exists, then $A[V(H) \cup \{v\}]$ is isomorphic to a one-subdivision of K_4 .

The auxiliary forms of the two-link theorem, the subdivision-attachment lemma, and the complete-bipartite path ranges are proved where they are used and are also formalized in Lean. Lemma 2.4 gives the remaining theta-residue calculation.

The proof uses the five published results above as stated. The Lean formalization proves internally the specializations used here: its rooted-path interfaces cover every value $2 \leq q \leq 4$ occurring below, and its two-connected branch applies the Sections 4–7 argument with empty deficiency rather than formalizing BGLP Theorem 1.3 verbatim.

2. Length arithmetic and simple-cycle assembly

The following lemmas contain all arithmetic used to combine path families.

Lemma 2.1 (sumsets of admissible progressions). Let

$$\mathcal{A} = \{a + i\alpha : 0 \leq i < s\}, \quad \mathcal{B} = \{b + j\beta : 0 \leq j < t\},$$

where $a, b \in \mathbb{Z}$, s, t are positive integers, and $\alpha, \beta \in \{1, 2\}$. Then $\mathcal{A} + \mathcal{B}$ contains an arithmetic progression of common difference one or two with at least $s + t - 1$ terms.

Proof. If $\alpha = \beta$, the entire sumset is

$$\{a + b + h\alpha : 0 \leq h \leq s + t - 2\}.$$

If $\{\alpha, \beta\} = \{1, 2\}$, translate and, if necessary, interchange the pairs $(\mathcal{A}, s)$ and $(\mathcal{B}, t)$ so that

$$\mathcal{A} = \{0, 1, \dots, s - 1\}, \quad \mathcal{B} = \{0, 2, \dots, 2(t - 1)\}.$$

If $s = 1$, the sum is the t -term progression $\mathcal{B}$. If $s \geq 2$, the intervals $[2j, 2j + s - 1]$, $0 \leq j < t$, overlap or are adjacent. Their union is therefore an interval of at least $s + t - 1$ integers. $\square$

Lemma 2.2 (concatenation). Let H be a subgraph of G , and let $X, Y \subseteq V(H)$ be disjoint. Let

$$\mathcal{A} = \{a_1, \dots, a_s\}, \quad \mathcal{B} = \{b_1, \dots, b_t\}$$

be increasing arithmetic progressions of positive integers, so that $a_i = a_1 + (i - 1)\alpha$ and $b_j = b_1 + (j - 1)\beta$ for $\alpha, \beta \in \{1, 2\}$, and suppose $b_1 \geq 2$. Assume that, for every $x \in X$, $y \in Y$, and $a \in \mathcal{A}$, the graph H has an x, y -path of length a . Assume also that, for every $b \in \mathcal{B}$, the graph G has an (X, Y) -path of length b whose internal vertices avoid $V(H)$. Then G has at least $s + t - 1$ admissible cycles.

The same conclusion holds with $X = Y$ if every outside path has distinct ends and, for every two distinct $x, y \in X$ and every $a \in \mathcal{A}$, the required x, y -path of length a exists in H .

We also use a two-variable form. Let $P_1, \dots, P_s$ and $Q_1, \dots, Q_t$ be path families with length sets $\mathcal{A}$ and $\mathcal{B}$, respectively. Suppose that, for every i, j , these paths and fixed connectors form a simple cycle C_{ij} , and that

$$|E(C_{ij})| = |E(P_i)| + |E(Q_j)| + \lambda$$

for a constant λ independent of i, j . Then the cycles C_{ij} contain an admissible subfamily of at least $s + t - 1$ members.

Proof. For each $b \in \mathcal{B}$, choose a corresponding outside path P_b , with ends x_b, y_b . Its internal vertices avoid $V(H)$, whereas every core x_b, y_b -path lies in H . The two paths meet exactly at x_b, y_b , and hence form a simple cycle whose length is their total length. Thus every element of $\mathcal{A} + \mathcal{B}$ occurs as a cycle length, and Lemma 2.1 gives the required admissible subfamily. The same argument applies when $X = Y$, because the outside path has distinct ends.

In the two-variable form, the cycle lengths contain $\lambda + \mathcal{A} + \mathcal{B}$, so Lemma 2.1 applies once more. $\square$

In each use of the two-variable form, we identify the supports of the variable paths and the fixed connectors. The stated intersection pattern then proves both simplicity and the displayed length formula. In particular, two three-term admissible families on such sides, with fixed connectors, yield five admissible cycles by Lemma 2.1. Their lengths include a multiple of five, since a five-term progression of difference one or two contains every residue modulo five.

Lemma 2.3 (the 3 + 3 lemma modulo five). Let $d \in \{1, 2\}$, let $a \in \mathbb{Z}$, and let $R \subseteq \mathbb{Z}/5\mathbb{Z}$ have three distinct elements. Then

$$\{a, a + d, a + 2d\} + R$$

contains 0 modulo five.

Proof. Put $R' = d^{-1}(a + R)$. Since $d \in \{1, 2\}$ is a unit modulo five, R' has three elements. The two three-element subsets R' and $\{0, -1, -2\}$ of $\mathbb{Z}/5\mathbb{Z}$ intersect. If $r' = -i$ lies in the intersection, with $i \in \{0, 1, 2\}$, then $a + id + r = 0$ for the corresponding $r \in R$. $\square$

Lemma 2.4 (theta residues in girth six). Let H be a theta in a graph of girth at least six. Then some two distinct vertices of H are joined in H by three paths whose lengths are pairwise distinct modulo five.

Proof. Let the three defining x, y -paths have lengths a, b, c . Each pair forms a cycle, so every pairwise sum is at least six. If the three lengths are already distinct modulo five, there is nothing to prove.

Suppose exactly two residues agree, say $a \equiv b \not\equiv c \pmod{5}$, and relabel so that $a \geq 3$. For $r \in \{1, 2\}$, let u_r be the vertex at distance r from x on the first path. The three u_r, y -paths that continue along the first path, or return to x and then follow the second or third path, have lengths

$$a - r, \quad r + b, \quad r + c \pmod{5}.$$

The first two residues are distinct, and so are the last two. The first and third can agree for at most one of $r = 1, 2$; the other value of r therefore gives three distinct residues.

It remains that $a \equiv b \equiv c \equiv t \pmod{5}$. Since every pairwise sum is at least six, relabel so that $a, b \geq 3$. Choose vertices at distances r and s from x on the first and second paths. The routes through x , through y , and through the third theta path have residues

$$r + s, \quad 2t - r - s, \quad r + 2t - s \pmod{5}.$$

For $t = 0, 3, 4$ take $(r, s) = (1, 1)$; for $t = 1$ take $(1, 2)$; and for $t = 2$ take $(2, 1)$. In each case the three residues are distinct. All selected vertices are internal, and distinct theta paths meet only at x, y , so every route used above is simple. $\square$

3. The rooted end-block reduction

Assume for a contradiction that a nonempty finite simple graph G_0 has minimum degree at least five and no cycle of length divisible by five. Choose a connected component G of G_0 . Every neighbor of a vertex of G also lies in G , so $\delta(G) \geq 5$; moreover, every cycle in G is a cycle in G_0 .

Write $n = |V(G)|$. If G is 2-connected, BGLP Theorem 1.3 with $k = 5$ gives five admissible cycles, one of whose lengths is divisible by five, unless

$$G \cong K_6 \quad \text{or} \quad G \cong K_{5,n-5}.$$

The first exception contains a 5-cycle. In the second, the minimum-degree hypothesis forces both parts to have at least five vertices, and hence gives a 10-cycle. Thus G is not 2-connected.

Choose an end block B of G , with cut vertex c . A leaf block exists by the block-cut tree. It is not a bridge, since the non-cut endpoint of an end bridge would have degree one in G . Thus B is 2-connected, and c has at least two neighbors in B . Every vertex of $B - c$ is inner in the end block, so all its incident edges in G lie in B . Therefore

$$d_B(v) \geq 5 \quad (v \in V(B) \setminus \{c\}). \quad (3.1)$$

If $d_B(c) \geq 5$, applying BGLP Theorem 1.3 to B again gives a cycle whose length is divisible by five: one of the five admissible cycles in the generic outcome has such a length, while the exceptional graphs contain a 5-cycle or a 10-cycle. Hence

$$2 \leq d_B(c) \leq 4. \quad (3.2)$$

Set

$$J = B - c, \quad D = \{v \in V(J) : d_J(v) = 4\}.$$

Deleting c removes at most one incident edge from each remaining vertex. Thus

$$d_J(v) = 4 \quad (v \in D), \quad d_J(v) \geq 5 \quad (v \in V(J) \setminus D), \quad (3.3a)$$

and every vertex of D is adjacent to c . In particular,

$$D \subseteq N_B(c), \quad |D| \leq d_B(c) \leq 4. \quad (3.3b)$$

For the rest of the proof, B and J satisfy (3.1)–(3.3b). Since $B \subseteq G \subseteq G_0$, the graph B contains no cycle of length divisible by five.

3.1. Two-connectivity after deleting the root

Lemma 3.1. The graph $J = B - c$ is 2-connected.

Proof. Since B is 2-connected, J is connected. Suppose that J is not 2-connected. Choose distinct end blocks E_1, E_2 of J , and let b_i be the cut vertex of J belonging to E_i . Neither end block is a bridge, because $\delta(J) \geq 4$.

For each i , an inner vertex $x_i \in V(E_i) \setminus \{b_i\}$ is adjacent to c ; otherwise b_i would separate $E_i - b_i$ from the rest of B . Every $v \in V(E_i) \setminus \{x_i, b_i\}$ is inner in E_i , and hence

$$d_{E_i}(v) = d_J(v) \geq 4.$$

Apply GHLM Theorem 3.1 with $q = 3$ to $E_i - x_i b_i$, rooted at x_i, b_i . The graph $(E_i - x_i b_i) + x_i b_i$ is 2-connected, and deleting the edge $x_i b_i$ changes only the degrees of the two roots. We obtain three admissible x_i, b_i -paths with lengths

$$\mathcal{A}_i = \{a_i, a_i + d_i, a_i + 2d_i\}, \quad d_i \in \{1, 2\}.$$

The block-cut tree supplies a b_1, b_2 -path R whose internal vertices avoid the two end-block interiors; if $b_1 = b_2$, take R to have length zero. Distinct end blocks have disjoint interiors and meet, if at all, only in their common cut vertex. For paths P_i chosen from the two families,

$$cx_1 + P_1 + R + \overleftarrow{P_2} + x_2c$$

is a simple cycle: the two end-block interiors are disjoint, R avoids both, and $c \notin V(J)$. When $b_1 = b_2$, the two sides share only c and their common endpoint. The possible lengths are

$$2 + |E(R)| + \mathcal{A}_1 + \mathcal{A}_2.$$

By Lemma 2.1 these include five admissible cycle lengths, one divisible by five. $\square$

3.2. Three-connectivity

Lemma 3.2. The graph J is 3-connected.

Proof. By Lemma 3.1, J is 2-connected, and (3.3a) gives $\delta(J) \geq 4$. Since $J \subseteq B$, it has no cycle whose length is divisible by five and hence no five admissible cycles. BGLP Lemma 2.3 with $k = 5$ and $r = 1$ gives

$$\kappa(J) \geq \frac{5+2}{2} - 1 = \frac{5}{2}.$$

Since vertex connectivity is integral, $\kappa(J) \geq 3$. $\square$

3.3. Root lifting

In every rooted auxiliary graph below, we shall verify the displayed degree bounds. After its construction-specific boundary losses are accounted for, only vertices of D may miss the required degree bound by one. The next lemma records the repair: when there are at least two such vertices, adjoining c restores every deficit and leaves c as the single COY exception.

Lemma 3.3 (root lifting). Let A be a subgraph of J , with $V(A) \subseteq V(J)$, and let $x, y \in V(A)$ be distinct roots such that $xy \notin E(A)$ and $A + xy$ is 2-connected. Let q be an integer with $2 \leq q \leq 4$, and let

$$Z \subseteq D \cap (V(A) \setminus \{x, y\})$$

satisfy

$$d_A(v) \geq q + 1 \quad (v \in V(A) \setminus (\{x, y\} \cup Z)), \quad d_A(z) \geq q \quad (z \in Z). \quad (3.4)$$

Assume additionally that $|V(A)| \geq 4$ in the case $|Z| = 1$. Then B contains q admissible x, y -paths whose internal vertices lie in $V(A)$, except that when $|Z| \geq 2$ they may also contain c .

Proof. If $Z = \emptyset$, apply GHLM Theorem 3.1 to A . If $|Z| = 1$, apply the COY theorem to A , taking the unique vertex of Z as the exception.

Now suppose $|Z| \geq 2$. Let A^+ be obtained from A by adjoining c and the edges cz for $z \in Z$. By (3.3b), each such edge belongs to B , so $A^+ \subseteq B$. The graph $A^+ + xy$ is 2-connected: it is obtained from $A + xy$ by adjoining a vertex with at least two neighbors. Each vertex of Z gains one degree, every other nonroot already has degree at least $q + 1$, and among the nonroots only c remains exceptional. Since $A + xy$ is 2-connected, $|V(A)| \geq 3$, and hence $|V(A^+)| \geq 4$. The COY theorem gives the required paths. $\square$

4. From 3-connectivity to 4-connectivity

We next exclude a three-vertex cut. The first lemma supplies the rooted 2-connectivity on either side of such a cut.

Lemma 4.1 (one component and two separator roots). Let $S = \{x, y, z\}$ be a three-vertex cut of J , let Q be a component of $J - S$, and put

$$A(Q; x, y) = J[Q \cup \{x, y\}] - xy.$$

Then $A(Q; x, y) + xy$ is 2-connected.

Proof. Each vertex of S has a neighbor in every component of $J - S$. By symmetry it suffices to consider z . If z had no neighbor in Q , then $J - \{x, y\}$ would separate Q from z and the remainder of the graph, contrary to Lemma 3.2. Hence Q together with either root is connected.

Let $K = A(Q; x, y) + xy$. Since $Q \neq \emptyset$ and $x, y \notin Q$, the graph K has at least three vertices; it is connected because Q is connected and each root has a neighbor in Q . Deleting either root leaves the other joined to Q . If some $w \in Q$ were a cut vertex of K , the edge xy would keep the roots in one component of $K - w$, while another component $R \subseteq Q - w$ would have no neighbor in $\{x, y\}$. Then $J - \{w, z\}$ would separate R from the rest of J , again contradicting 3-connectivity. Thus K is 2-connected. $\square$

Every component Q of $J - S$ has at least two vertices, since a singleton would have degree at most three. We shall also use the following extension. If z has at least two neighbors in Q , then

$$(J[Q \cup S] - xy) + xy$$

is 2-connected: start with $A(Q; x, y) + xy$ and adjoin z through two of its neighbors in Q .

For later use, if Q is any component of $J - S$ and $A = A(Q; x, y)$, then the only separator vertex omitted from A is z . Deleting xy does not affect a vertex of Q , so

$$d_A(v) \geq 4 \quad (v \in Q \setminus D), \quad d_A(v) \geq 3 \quad (v \in Q \cap D). \quad (4.1)$$

Lemma 4.2. If S is a three-vertex cut of J , then every component of $J - S$ contains at least two vertices of D . Consequently, $J - S$ has exactly two components, $|D| = 4$, and each component contains exactly two vertices of D .

Proof. Write $S = \{x, y, z\}$. Suppose that a component Q_0 of $J - S$ contains at most one vertex of D , and let Q_1 be another component. For $i \in \{0, 1\}$, set

$$A_i = A(Q_i; x, y).$$

Let $Z_i = D \cap Q_i$. Since $|Q_i| \geq 2$, Lemma 3.3 with $q = 3$ applies and gives three admissible x, y -paths on each side: Lemma 4.1 gives the augmented 2-connectivity, (4.1) gives the degrees, and $|V(A_i)| = |Q_i| + 2 \geq 4$ supplies the exceptional-case order. The Q_0 -family avoids c , because $|Z_0| \leq 1$; the Q_1 -family may use c .

Apart from the roots, the two families lie in the disjoint components Q_0, Q_1 , and at most one may contain c . Any two selected paths therefore form a simple cycle. Lemma 2.1 gives five admissible cycle lengths, a contradiction. Thus every component of $J - S$ contains at least two vertices of D . Since $J - S$ has at least two components and $|D| \leq 4$, it has exactly two; each contains exactly two vertices of D , and $D \cap S = \emptyset$. $\square$

Lemma 4.3. The graph J is 4-connected.

Proof. Suppose that S is a three-vertex cut. Lemma 4.2 shows that $J - S$ has precisely two components Q_1, Q_2 and gives

$$|D \cap Q_1| = |D \cap Q_2| = 2, \quad D \cap S = \emptyset. \quad (4.2)$$

Every vertex of S therefore has degree at least five in J . Retain one separator vertex, say z , and denote the other two by x, y . The vertex z has at most two neighbors in S , so one of Q_1, Q_2 contains at least two of its neighbors; relabel that component as Q_2 .

On the first side, put

$$A_1 = J[Q_1 \cup \{x, y\}] - xy, \quad Z_1 = D \cap Q_1.$$

Lemma 4.1 and (4.1) allow Lemma 3.3 with $q = 3$. Since $|Z_1| = 2$, it yields three admissible x, y -paths with interiors in $Q_1 \cup \{c\}$.

On the second side, put

$$A_2 = J[Q_2 \cup S] - xy.$$

By the extension following Lemma 4.1, $A_2 + xy$ is 2-connected. Every vertex of Q_2 retains all its J -neighbors in $Q_2 \cup S$, and therefore has degree at least four in A_2 . The only other vertices are the roots x, y and the exceptional vertex z . Since $|V(A_2)| \geq 5$, the COY theorem with $q = 3$ gives three admissible x, y -paths with interiors in $Q_2 \cup \{z\}$.

The two families meet only at x, y : their remaining vertices lie on opposite sides of the cut, and only the first family can contain c . Pairwise unions are therefore simple cycles. Lemma 2.1 gives five admissible lengths, a contradiction. Hence no three-cut exists. Since $\delta(J) \geq 4$, the graph has at least five vertices, and Lemma 3.2 already excludes cuts of size at most two. Thus J is 4-connected. $\square$

5. First local exclusions: K_4^- and triangles

5.1. Excluding K_4^-

Lemma 5.1. The graph J contains no subgraph isomorphic to K_4^- .

Proof. Suppose that distinct vertices x, y, a, b contain the five edges

$$xy, xa, ya, yb, ab;$$

no condition is imposed on xb . No vertex $w \in V(J) \setminus \{x, y, a, b\}$ is adjacent to both a and b , since then

$$x - y - b - w - a - x$$

would be a 5-cycle. Put $K = \{a, b\}$.

Deleting two vertices from the 4-connected graph J leaves a 2-connected graph, so $J - K$ is 2-connected. Let

$$A = (J - K) - xy,$$

with roots x, y . Adding xy gives a 2-connected graph. By the preceding paragraph, every vertex of $V(A) \setminus \{x, y\}$ has at most one neighbor in K , and hence

$$d_A(v) \geq 4 \quad (v \in V(A) \setminus (D \cup \{x, y\})), \quad d_A(v) \geq 3 \quad (v \in (D \cap V(A)) \setminus \{x, y\}). \quad (5.1)$$

Set $Z = D \cap (V(A) \setminus \{x, y\})$. If $|V(A)| = 3$, then $|V(J)| = 5$, and 4-connectivity forces $J = K_5$, which contains a 5-cycle. Since $A + xy$ is 2-connected, $|V(A)| \geq 3$; the equality case has just been excluded. Thus $|V(A)| \geq 4$, and Lemma 3.3 with $q = 3$ gives three admissible x, y -paths whose interiors avoid K , apart from the possible vertex c .

The subgraph on $K \cup \{x, y\}$ contains x, y -paths of lengths 1, 2, 3:

$$xy, \quad xay, \quad xaby.$$

Each of these core paths meets every lifted path only at x, y . Lemma 2.2 therefore gives five admissible simple cycles in B , a contradiction. $\square$

5.2. Excluding triangles

Contraction fact. Let G be a 4-connected simple graph and let $qr \in E(G)$. After contracting qr and suppressing loops and parallel edges, the resulting simple graph G/qr is 3-connected.

Proof. Let

$$\pi : V(G) \longrightarrow V(G/qr)$$

be the contraction map. Since G has at least five vertices, G/qr has at least four. Let $X \subseteq V(G/qr)$ with $|X| \leq 2$, and put $\tilde{X} = \pi^{-1}(X)$. If the contracted vertex is not in X , then $|\tilde{X}| = |X|$; if it is, then $|\tilde{X}| = |X| + 1$. Thus $|\tilde{X}| \leq 3$, and $G - \tilde{X}$ is connected.

Take $x, y \in V(G/qr) \setminus X$ and choose preimages $\tilde{x}, \tilde{y} \notin \tilde{X}$. Apply π to the vertex sequence of a $\tilde{x}, \tilde{y}$ -path in $G - \tilde{X}$ and delete consecutive repetitions. Consecutive distinct images are adjacent in the simple quotient, so the resulting sequence is an x, y -walk in $(G/qr) - X$. Every walk contains a path with the same endpoints. Hence $(G/qr) - X$ is connected for every $|X| \leq 2$, and G/qr is 3-connected. $\square$

Lemma 5.2. The graph J is triangle-free.

Proof. Suppose that pqr is a triangle. By Lemma 5.1, every vertex of $J - \{p, q, r\}$ has at most one neighbor in this triangle.

Choose an edge of the triangle, say qr , whose contraction leaves c with at least two distinct neighbors. Contracting one edge can identify at most two neighbors of c , so any edge works when $d_B(c) \geq 3$. When $d_B(c) = 2$, only the edge joining those two neighbors can identify them, and at most one triangle edge does so.

Contract qr to a vertex t in B , suppress loops and parallel edges, and delete pt . Let the resulting graph be L , rooted at p, t . The graph J/qr is 3-connected and hence 2-connected. By the choice of qr , the vertex c has two neighbors in J/qr ; adjoining it through those neighbors shows that B/qr is 2-connected, and

$$L + pt = B/qr.$$

For $v \in V(B) \setminus \{p, q, r, c\}$, identify v with its unchanged image in L . The contraction merges no two edges incident with v , since v has at most one neighbor in the triangle. Hence

$$d_L(v) = d_B(v) \geq 5 \quad (v \in V(L) \setminus \{p, t, c\}).$$

Moreover $|V(L)| \geq 5$: the 4-connected graph J has at least five vertices, B has the additional vertex c , and the contraction removes one vertex. The COY theorem with $q = 4$ and exceptional vertex c gives four admissible p, t -paths in L .

The edge pt is absent, so each rooted path has length at least two. In a simple p, t -path, t occurs only at the end. Every other quotient vertex has a unique unchanged preimage, and every preceding path edge is an edge of B . If v is the penultimate vertex, then vt comes from vq or vr in B ; replace it by one such edge. No earlier vertex represents q or r , so this gives a simple path of the same length ending at one of q, r and avoiding the other.

Close a lift ending at q by qp or by qrp , and use the symmetric closures for a lift ending at r . The unused triangle vertex is absent from the lift, so each closing route meets it only at its endpoints; both unions are simple cycles and add one or two edges. If the four path lengths are $a, a + d, a + 2d, a + 3d$, the closure lengths are

$$\{a + id + j : 0 \leq i \leq 3, j \in \{1, 2\}\}.$$

For $d = 1$ these contain $a + 1, \dots, a + 5$, and for $d = 2$ they contain $a + 1, \dots, a + 8$. In either case one length is divisible by five, a contradiction. $\square$

6. Boundary lifting and the four-cycle exclusion

To exclude a four-cycle, we enlarge a complete bipartite core as far as possible and construct admissible paths through a component outside it. The next two lemmas justify the recurring construction: replace attachments to the core by two artificial roots, apply a rooted path theorem, and then replace the artificial boundary edges by genuine edges back into the core.

6.1. Boundary-root lifting

Lemma 6.1. Let H have distinct distinguished vertices x, y , and put $Q = H - \{x, y\}$. Suppose that:

1. Q is connected;
2. H contains independent edges xu and yv with distinct $u, v \in V(Q)$; and
3. every end block of Q has an inner vertex adjacent in H to x or y .

Then $H + xy$ is 2-connected.

Proof. The graph $H + xy$ is connected, and deleting either root leaves the other joined to Q . The independent boundary edges exhibit the four distinct vertices x, y, u, v , so $|V(H)| \geq 4$ and the order requirement for 2-connectivity holds. Suppose that $w \in Q$ is a cut vertex of $H + xy$. The edge xy keeps the roots together, and $u \neq v$ ensures that at least one of their two attachments survives the deletion of w . Thus the roots lie with one component of $Q - w$, while some other component contains no neighbor of either root. The block-cut tree supplies an end block of Q in that component and away from w , contradicting (3). $\square$

Lemma 6.2 (boundary-root lifting). Let $c \in V(B)$, put $J = B - c$, and let $D \subseteq V(J)$ satisfy $D \subseteq N_B(c)$. Let H have distinct roots α, β , with $\alpha\beta \notin E(H)$ and $c \notin V(H)$, and put

$$Q = H - \{\alpha, \beta\}.$$

Suppose that Q admits an injective graph embedding into J . Relabel the old vertices by their images under this embedding; henceforth $V(Q) \subseteq V(J)$ and $Q \subseteq J$. Suppose also that $H + \alpha\beta$ is 2-connected. Set $Z = D \cap V(Q)$. Let q be an integer with $2 \leq q \leq 4$, and suppose

$$d_H(v) \geq q + 1 \quad (v \in V(Q) \setminus Z), \quad d_H(z) \geq q \quad (z \in Z). \quad (6.1)$$

Assume additionally that $|V(H)| \geq 4$ in the case $|Z| = 1$. Then:

1. if $|Z| \leq 1$, H contains q admissible α, β -paths whose internal vertices lie in Q ;
2. if $|Z| \geq 2$, the graph H^+ obtained by adjoining c and precisely the edges cz , $z \in Z$, contains q admissible α, β -paths whose internal vertices lie in $Q \cup \{c\}$.

Suppose in addition that $\alpha, \beta \notin V(B)$, and let $F \subseteq J$ satisfy $V(F) \cap V(Q) = \emptyset$. For each edge $\alpha u \in E(H)$, choose a nonempty set

$$L(u) \subseteq N_J(u) \cap V(F),$$

and for each edge $v\beta \in E(H)$ choose a nonempty set

$$R(v) \subseteq N_J(v) \cap V(F).$$

If, for every rooted path supplied above with first old vertex u and last old vertex v , one can choose $x \in L(u)$ and $y \in R(v)$ with $x \neq y$, then replacing αu by xu and $v\beta$ by vy gives an x, y -path in B of the same length. Its internal vertices lie in $Q \cup \{c\}$ and avoid F . The endpoint pair may depend on the path, but the length sequence remains admissible.

Proof. For $Z = \emptyset$, apply GHLM Theorem 3.1 to H . For $|Z| = 1$, apply the COY theorem to H , with the unique vertex of Z as the exception. If $|Z| \geq 2$, then $H^+ + \alpha\beta$ is 2-connected because it is obtained from $H + \alpha\beta$ by adjoining c through at least two neighbors. Each vertex of Z gains one degree, every other nonroot already has degree at least $q + 1$, and only c is exceptional. The two roots, two distinct vertices of Z , and c give $|V(H^+)| \geq 5$, so the COY order condition also holds. The theorem gives the asserted paths.

For the boundary replacement, let P be one of these simple rooted paths, and let u, v be the neighbors on P of α, β , respectively. Deleting the artificial roots leaves a simple u, v -path, reduced to the single vertex u when $u = v$. Choose distinct $x \in L(u)$ and $y \in R(v)$. Because F is disjoint from Q and $c \notin V(J)$, neither endpoint lies on the remaining segment. Replacing αu and $v\beta$ by xu and vy therefore preserves both simplicity and length. $\square$

6.2. A lifted complete-bipartite core

Let $K_{s,t}^-$ denote a graph obtained from $K_{s,t}$ by deleting one edge, and set

$$\mathcal{F} = \{K_{s,t} : \min\{s, t\} \geq 2\} \cup \{K_{s,t}^- : \min\{s, t\} \geq 2, \max\{s, t\} \geq 3\}.$$

We shall use the following elementary path ranges.

Complete-bipartite path lemma. Let $F = K_{s,t}$ with $r = \min\{s, t\} \geq 2$, or let $F = K_{s,t}^-$ with $r \geq 3$. Let (S, T) be its bipartition.

- If x, y lie in opposite parts and $xy \in E(F)$, then F has an x, y -path of every odd length $1, 3, \dots, 2r - 1$.
- If x, y are distinct vertices in the same part, then F has an x, y -path of every even length $2, 4, \dots, 2r - 2$. If that part has more than r vertices, it also has one of length $2r$.

Proof. The complete case follows by choosing the required number of vertices from each part and alternating. Suppose that the missing edge is ab , where $a \in S$ and $b \in T$; exchanging S, T handles the reverse orientations below.

First let $x, y \in S$ and let the desired length be $2m \leq 2r - 2$. Choose distinct vertices

$$s_0 = x, s_1, \dots, s_{m-1}, s_m = y \quad \text{and} \quad t_0, \dots, t_{m-1},$$

with every $t_i \neq b$. This is possible because $|T \setminus \{b\}| \geq r - 1 \geq m$, and the alternating sequence

$$s_0 t_0 s_1 t_1 \cdots s_{m-1} t_{m-1} s_m$$

is the required path.

For the terminal same-part length $2r$, assume that the endpoint part, say S , has more than r vertices. Use an alternating sequence

$$s_0 t_0 s_1 t_1 \cdots s_{r-1} t_{r-1} s_r, \quad s_0 = x, \quad s_r = y.$$

Avoid a or b , the endpoints of the deleted edge, whenever the vertex in question is not one of the prescribed path endpoints and enough vertices remain in its part. If both a and b must occur, place b at t_1 when $a = x$, at t_0 when $a = y$, and otherwise place a at s_1 and b at t_2 . Since $r \geq 3$, these positions exist, and in each case a and b are nonconsecutive.

Finally let $x \in S, y \in T$, and let the desired length be $2m + 1$. The case $m = 0$ is the assumed edge xy . Otherwise choose an alternating sequence

$$s_0 t_0 s_1 t_1 \cdots s_m t_m, \quad s_0 = x, \quad t_m = y.$$

Whenever possible, omit an unprescribed one of a, b . If both are forced to occur, then $m \geq 2$ and they can be placed as follows: if $a = x$, put $b = t_1$; if $b = y$, put $a = s_1$; and if neither is prescribed, put $a = s_m$ and $b = t_0$. The endpoints cannot satisfy both $a = x$ and $b = y$ because xy is present. Again the missing pair is nonconsecutive, so every displayed sequence is a path in F . $\square$

Lemma 6.3 (lifted core lemma). Under the standing setup $J = B - c$ and $D = \{v \in V(J) : d_J(v) = 4\}$, assume

$$\kappa(J) \geq 4, \quad J \text{ is triangle-free}, \quad \delta(J) \geq 4, \quad D \subseteq N_B(c), \quad |D| \leq 4.$$

Suppose that $F \subseteq J$ is isomorphic to a graph in $\mathcal{F}$, with bipartition (S, T) , and put $r = \min\{|S|, |T|\}$. If every vertex of $J - V(F)$ has at most $r - 1$ neighbors in F , then B contains a cycle of length divisible by five.

Proof. Suppose that B has no cycle of length divisible by five. Let $\mathcal{C}$ be the nonempty family of cores $F' \in \mathcal{F}$ for which every exterior vertex has at most $r(F') - 1$ neighbors in F' , where $r(F')$ is its smaller part size. Choose $F \in \mathcal{C}$ of minimum order, label its parts (S, T) , and reset $r = |S| \leq |T|$.

Triangle-freeness forbids edges within either part: for any same-part edge, an opposite-part vertex not involved in the possible missing edge completes a triangle. Thus the induced graph on the same vertex set is either F or the complete bipartite graph obtained by restoring the one missing edge. Replacing F by that induced graph preserves both its order and the outside-neighbor bound. We may therefore assume that F is induced.

If $r \geq 5$, then F contains a 10-cycle. This is immediate for a complete core. If the missing edge is $s_1 t_1$, choose five vertices in each part and use

$$s_1 t_2 s_2 t_1 s_3 t_3 s_4 t_4 s_5 t_5 s_1.$$

Hence

$$2 \leq r \leq 4. \tag{6.2}$$

The core is not spanning. If $J = F = K_{s,t}^-$, then $\delta(J) \geq 4$ forces $r \geq 5$. If $J = F = K_{s,t}$, then (6.2) and $\delta(J) \geq 4$ force $r = 4$; but then more than four vertices have degree four, contrary to $|D| \leq 4$.

Fix a component Q of $J - V(F)$. It has at least two vertices, since a singleton would have degree at most $r - 1 \leq 3$. Every end block of Q has an inner vertex adjacent to F . If Q has no cut vertex (including the case $Q \cong K_2$), then Q is its unique end block, and the assertion follows from the connectedness of J . Otherwise, if an end block E had no such inner attachment and b were its unique cut vertex, then $J - b$ would separate $V(E) \setminus \{b\}$ from the rest of J , contradicting 4-connectivity. This supplies the end-block condition in Lemma 6.1 whenever all core attachments are represented; the one construction that omits a boundary vertex is checked separately below. Every auxiliary graph has at least four vertices, so the order condition in Lemma 6.2 causes no further issue.

In every case below, an auxiliary graph retains all edges of Q , deletes the boundary edges from Q to the core, and replaces all boundary edges of each represented type at a vertex by one edge to the corresponding artificial root.

Each case now follows the same pattern. Core attachments are represented by two artificial roots; the outside-neighbor bound gives the degree estimate; Lemma 6.2 returns admissible paths to genuine, distinct endpoints of F ; and the bipartite path ranges close them through the core. It remains to distinguish whether Q has attachments in one or both parts.

The numerical bookkeeping is the same in every case. The middle columns count available lengths, and the last records the admissible-term guarantee from Lemma 2.1:

case	outside count	core count	guarantee
$r = 2$, Q has attachments in both parts	4	2	$4 + 2 - 1 = 5$
$r = 2$, Q has attachments in one part	4	2	$4 + 2 - 1 = 5$
$3 \leq r \leq 4$, Q has attachments in both parts	$6 - r$	r	$(6 - r) + r - 1 = 5$
$3 \leq r \leq 4$, Q has attachments in one part	$7 - r$	$r - 1$	$(7 - r) + (r - 1) - 1 = 5$

Rank two Suppose $r = 2$. If $F = K_{2,t}^-$, let $x_1 y_1$ be its missing edge, with $x_1 \in S$ and $y_1 \in T$. Deleting y_1 leaves a smaller $K_{2,t-1}$ in $\mathcal{F}$, since $t \geq 3$, with the same outside-neighbor bound: the newly external vertex y_1 has exactly one neighbor in the smaller core, and shrinking the core cannot increase any other exterior count. This contradicts the minimality of F . Thus $F = K_{2,t}$.¹

Since J is 3-connected and $|S| = 2$, the component Q must have an attachment in T . Suppose that it also has an attachment in S . Form H_{ST} from Q by adjoining roots s^*, t^* and joining $v \in Q$ to a root whenever v has a neighbor in the represented part. The outside-neighbor bound is one, so the two root attachments have distinct endpoints in Q . Lemma 6.1 gives that $H_{ST} + s^* t^*$ is 2-connected. Each boundary edge deleted at a vertex of Q is replaced by the corresponding root edge, and hence

$$d_{H_{ST}}(v) = d_J(v) \geq 5 \quad (v \in V(Q) \setminus D), \quad d_{H_{ST}}(v) = 4 \quad (v \in V(Q) \cap D).$$

For an edge $s^* v$, use $N_J(v) \cap S$ as its represented endpoint set; for an edge vt^* , use $N_J(v) \cap T$. The represented endpoints lie in opposite parts and are therefore distinct. Lemma 6.2 with $q = 4$ gives four admissible outside paths, while the core gives lengths one and three between every corresponding pair of endpoints. Lemma 2.2 yields five admissible cycles.

We may therefore assume that every attachment of Q lies in T . Three-connectivity gives at least three attachment vertices in T , so $|T| \geq 3$. The bipartite attachment graph between Q and T has a matching of size two. Otherwise a bipartite graph with no two independent edges is a star, so all attachment edges share one endpoint. If that endpoint lies in T , deleting it separates Q ; if it is a vertex $v \in Q$, deleting v separates the nonempty set $Q - \{v\}$ from the core. Choose independent attachment edges with distinct ends $y_1, y_2 \in T$, and partition $T = T_1 \cup T_2$ with $y_i \in T_i$.

Form H_{12} by adjoining roots t_1^*, t_2^* and joining $v \in Q$ to t_i^* when v has a neighbor in T_i . Lemma 6.1 applies, and the same degree calculation gives $d_{H_{12}}(v) = d_J(v)$. Represent the two boundary edges by the corresponding neighborhoods in T_1 and T_2 . Lemma 6.2 with $q = 4$ gives four admissible outside paths. Between any endpoint in T_1 and any endpoint in T_2 , the core $K_{2,t}$ has paths of lengths two and four; the latter uses a third vertex of T . Lemma 2.2 again gives five admissible cycles. Both rank-two cases are impossible, so

$$r \geq 3. \tag{6.3}$$

Attachments in both parts Suppose that Q has attachments in both S and T . First let $F = K_{s,t}$. Form H_{opp} by adjoining roots that represent S and T . No vertex of Q has neighbors in both parts, since any such pair of neighbors would form a triangle with their cross-edge. Thus the root attachments are independent, and Lemma 6.1 applies. For $v \in Q$, put $m = |N_J(v) \cap V(F)| \leq r - 1$. If $m > 0$, replacing the m core edges by one root edge gives degree $d_J(v) - m + 1$; if $m = 0$, the degree is unchanged. Hence

$$d_{H_{\text{opp}}}(v) \geq d_J(v) - r + 2 \geq \begin{cases} 7 - r, & v \in V(Q) \setminus D, \\ 6 - r, & v \in V(Q) \cap D. \end{cases} \tag{6.4}$$

Use the original neighborhoods in S and T as the represented endpoint sets. Lemma 6.2 with

$$q = 6 - r \in \{2, 3\}$$

¹This is also the rank-two reduction used in the complete-bipartite step of the proof of BGLP Theorem 1.3; every subsequent deleted-edge use in that proof has smaller part at least three.

gives $6 - r$ admissible outside paths between the two parts. The core supplies the r odd lengths $1, 3, \dots, 2r - 1$ between every endpoint pair. Since $(6 - r) + r - 1 = 5$, Lemma 2.2 gives a contradiction.

Now let $F = K_{s,t}^-$, with missing edge $x_1 y_1$, where $x_1 \in S$ and $y_1 \in T$. Some core endpoint of an attachment edge from Q lies outside $\{x_1, y_1\}$. Otherwise $J - \{x_1, y_1\}$ would separate Q from the nonempty graph $F - \{x_1, y_1\}$. Interchanging S, T if necessary, assume that Q has a neighbor in $T \setminus \{y_1\}$; by the present case assumption, it also has a neighbor in S . Form H_{del} with roots representing S and $T \setminus \{y_1\}$.

Every end block E of Q has an inner vertex adjacent to $F - \{y_1\}$. If Q has no cut vertex (including $Q \cong K_2$), then it is its unique end block, and the chosen attachment in $T \setminus \{y_1\}$ supplies such a vertex. Otherwise, let b be the unique cut vertex of an end block E . If $E - b$ had no neighbor in $F - \{y_1\}$, then no edge of $J - \{b, y_1\}$ would leave $E - b$: inside Q , every such edge passes through b , while by assumption its only possible core attachment is y_1 . Thus $\{b, y_1\}$ would separate J , contrary to 4-connectivity.

Finally, no vertex of Q is adjacent both to a vertex of S and to a vertex of $T \setminus \{y_1\}$, since those two core vertices are adjacent and would form a triangle with it. Hence the two root attachments have distinct ends in Q . Lemma 6.1 applies.

For $v \in Q$, put $m = |N_J(v) \cap V(F)| \leq r - 1$. If v has a represented neighbor, the new graph adds a root edge and

$$d_{H_{\text{del}}}(v) \geq d_J(v) - m + 1 \geq d_J(v) - r + 2.$$

If it has none, then all its core neighbors lie in $\{y_1\}$, so $d_{H_{\text{del}}}(v) \geq d_J(v) - 1 \geq d_J(v) - r + 2$. Thus the bounds in (6.4) also hold for H_{del} . Lemma 6.2 supplies $6 - r$ admissible paths from S to $T \setminus \{y_1\}$. Every resulting endpoint pair is joined by an edge of F , and the complete-bipartite path lemma applies because $r \geq 3$; it supplies the r odd lengths

$$1, 3, \dots, 2r - 1.$$

Since $(6 - r) + r - 1 = 5$, Lemma 2.2 gives five admissible cycles. We conclude that Q has neighbors in only one part of F .

Attachments in one part By symmetry, assume that every attachment of Q lies in T . The attachment graph between Q and T has a matching of size two; otherwise it is a star. Deleting its center disconnects all of Q when the center lies in T , or the nonempty set $Q - \{v\}$ when the center is $v \in Q$. Choose independent attachment edges with distinct ends $y_1, y_2 \in T$, and partition $T = T_1 \cup T_2$ with $y_i \in T_i$.

Construct H_{same} from Q by adjoining roots t_1^*, t_2^* . For $v \in Q$, add

$$vt_1^* \quad \text{if } d_{T_1}(v) \geq 1 \text{ or } d_T(v) \geq 2,$$

and

$$vt_2^* \quad \text{if } d_{T_2}(v) \geq 1 \text{ or } d_T(v) \geq 2.$$

The chosen matching gives independent root attachments, and every end-block attachment is represented; hence Lemma 6.1 gives that $H_{\text{same}} + t_1^* t_2^*$ is 2-connected. If $d_T(v) \leq 1$, every deleted boundary edge is replaced. If $d_T(v) \geq 2$, two root edges are added and $d_T(v) \leq r - 1$. Hence

$$d_{H_{\text{same}}}(v) \geq d_J(v) - r + 3 \geq \begin{cases} 8 - r, & v \in V(Q) \setminus D, \\ 7 - r, & v \in V(Q) \cap D. \end{cases} \quad (6.5)$$

For each root edge vt_i^* , define its represented endpoint set by

$$L_i(v) = \begin{cases} N_T(v), & d_T(v) \geq 2, \\ N_T(v) \cap T_i, & d_T(v) = 1. \end{cases}$$

This set is nonempty whenever vt_i^* is present. We claim that, for every rooted t_1^*, t_2^* -path, the two represented endpoints can be chosen distinct.

Let u and v be the first and last old vertices of the rooted path. If $u = v$, then the presence of both root edges forces $d_T(u) \geq 2$: a unique T -neighbor belongs to only one of T_1, T_2 . We may therefore choose two distinct neighbors of u . If $u \neq v$ and both represented sets are singletons, their elements lie in the disjoint sets T_1, T_2 . If at least one represented set has at least two elements, choose the endpoint from the other set first and then avoid it in the larger set. This proves the boundary-replacement condition in Lemma 6.2.

Lemma 6.2 with

$$q = 7 - r \in \{3, 4\}$$

therefore gives $7 - r$ admissible paths with distinct ends in T and interiors in $Q \cup \{c\}$. Between any two distinct vertices of T , the core supplies the $r - 1$ even lengths

$$2, 4, \dots, 2r - 2.$$

Since $(7 - r) + (r - 1) - 1 = 5$, Lemma 2.2 gives five admissible cycles. This final case is also impossible, proving the lemma. $\square$

6.3. Excluding four-cycles

Lemma 6.4. The graph J contains no four-cycle.

Proof. A four-cycle contains a $K_{2,2}$. Let $s \geq 2$ be maximal such that $K_{s,s} \subseteq J$. Since $K_{5,5}$ contains a 10-cycle, $s \leq 4$.

Suppose first that J contains $K_{s+1,s+1}^-$ with bipartition (S, T) . Maximality of s excludes $K_{s+1,s+1}$, so its designated missing edge $x_0 y_0$ is genuinely absent. An outside vertex has at most s neighbors in either part: a vertex adjacent to all of S , for example, could replace y_0 and complete a $K_{s+1,s+1}$. Moreover, a vertex meeting both parts can meet only x_0 and y_0 , since any other cross-pair is an edge of the core and would complete a triangle. Thus every outside vertex has at most s neighbors in the whole core. Lemma 6.3, with $r = s + 1$, gives a contradiction. Hence J contains no $K_{s+1,s+1}^-$.

Suppose next that $K_{s,s+1} \subseteq J$. Fix its s -vertex side S , and choose a largest set $T \subseteq V(J) \setminus S$ such that all edges between S and T are present. Put $t = |T|$, and let F be the resulting $K_{s,t}$. Triangle-freeness prevents an outside vertex from meeting both parts. Maximality of t prevents it from meeting all of S . If an outside vertex v had s neighbors $T_0 \subseteq T$, choose $y \in T \setminus T_0$, possible because $t \geq s + 1$. Then the bipartition

$$(S \cup \{v\}, T_0 \cup \{y\})$$

has every required cross-edge except possibly vy . If vy is absent, these vertices contain a $K_{s+1,s+1}^-$, already excluded; if it is present, they contain a $K_{s+1,s+1}$, contradicting the maximality of s . Thus every outside vertex has at most $s - 1$ neighbors in F , and Lemma 6.3 gives a contradiction. Hence J contains no $K_{s,s+1}$.

Finally, take $F = K_{s,s}$. An outside vertex meets at most one part and cannot meet all s vertices there, since that would create a $K_{s,s+1}$. Lemma 6.3 applies once more and gives the final contradiction. $\square$

The structural reduction is now complete. Lemmas 5.2 and 6.4 exclude cycles of lengths three and four, while the counterexample assumption excludes 5- and 10-cycles in $J \subseteq B$. Therefore

$$\text{girth}(J) \geq 6, \quad J \text{ has no 10-cycle.} \quad (6.6)$$

7. The minimum-theta argument

We now use (6.6) for the final contradiction. Since J is 2-connected and $\delta(J) \geq 4$, it is not a cycle and therefore contains a theta. Choose one, H , of minimum order. By GHLM Lemma 5.10, H is induced, at most one vertex outside H has two or more neighbors in H , and adjoining that vertex, if it exists, produces an induced one-subdivision of K_4 .

Subdivision attachment lemma. Let K be a one-subdivision of K_4 in a graph A of girth at least five. If $v \in V(A) \setminus V(K)$ has two distinct neighbors in K , then A contains a cycle of length five or ten.

Proof. Write a_1, a_2, a_3, a_4 for the branch vertices of K , and let $z_{ij} = z_{ji}$ be the subdivision vertex on the edge $a_i a_j$ of the underlying K_4 . Let x, y be two distinct neighbors of v in K .

If both are branch vertices, say $x = a_i$ and $y = a_j$, then

$$v a_i z_{ij} a_j v$$

is a 4-cycle, contrary to the girth assumption.

Suppose next that $x = a_i$ is a branch vertex and $y = z_{jk}$ is a subdivision vertex. If $i \in \{j, k\}$, then x and y are adjacent and form a triangle with v . Otherwise i, j, k are distinct, and

$$v a_i z_{ij} a_j z_{jk} v$$

is a 5-cycle.

It remains that $x = z_{ij}$ and $y = z_{k\ell}$ are subdivision vertices. If the corresponding edges of K_4 share an endpoint, then x and y are joined by a two-edge path through that branch vertex, giving a 4-cycle with v . Otherwise the two edges are disjoint. Relabeling, take $x = z_{12}$ and $y = z_{34}$. Then

$$z_{12} a_1 z_{13} a_3 z_{23} a_2 z_{24} a_4 z_{34}$$

is an eight-edge path in K , and adjoining the two edges from its ends to v gives a 10-cycle. $\square$

Define $T = H$ if no such vertex exists; otherwise let T be the induced one-subdivision of K_4 obtained by adjoining the unique such vertex. For $Q \subseteq V(J) \setminus V(T)$, write

$$N_T(Q) = N_J(Q) \cap V(T).$$

If no vertex outside H has two neighbors in H , then $T = H$, and GHLM Lemma 5.10 already gives

$$|N_T(v)| \leq 1 \quad (v \in V(J) \setminus V(T)).$$

Otherwise T is a one-subdivision of K_4 . In that case, a vertex outside T with two neighbors in T would, by the subdivision attachment lemma, create a 5-cycle or a 10-cycle, contrary to (6.6). Consequently,

$$T \text{ is induced and 2-connected,} \quad \Delta(T) \leq 3, \quad |N_T(v)| \leq 1 \quad (v \in V(J) \setminus V(T)). \quad (7.1)$$

By Lemma 2.4, there are distinct $x, y \in V(H) \subseteq V(T)$ joined in T by three paths whose lengths are pairwise distinct modulo five.

Set $M = J - V(T)$. The graph M is nonempty, since otherwise $J = T$ would have maximum degree at most three. By (3.3a) and (7.1), each vertex loses at most one neighbor when T is deleted, so

$$d_M(v) \geq 4 \quad (v \in V(M) \setminus D), \quad d_M(v) \geq 3 \quad (v \in V(M) \cap D). \quad (7.2)$$

We show first that every component of M is 2-connected and then that M has only one component.

7.1. The components of the exterior

Lemma 7.1. Every component of M is 2-connected.

Proof. Suppose that a component Q is not 2-connected. Since Q is a component of M , (7.2) gives minimum degree at least three in Q ; hence every end block is nontrivial. Choose distinct end blocks E_1, E_2 , and let b_i be the cut vertex of Q belonging to E_i .

For each i ,

$$|N_T(E_i - b_i)| \geq 3. \quad (7.3)$$

Indeed, if this set had at most two vertices, deleting it together with b_i would separate the nonempty interior $E_i - b_i$ from the rest of J . No edge leaves that interior inside Q except through b_i , and at least one vertex of the 2-connected graph T survives. This contradicts the 4-connectivity of J .

Choose attachment edges $u_i y_i$ with $u_i \in E_i - b_i$ and $y_i \in T$, with $y_1 \neq y_2$; the three choices available in (7.3) permit this. Distinct end blocks have disjoint interiors, although their cut vertices may coincide. The block-cut tree gives a b_1, b_2 -path

$$R \subseteq Q - ((E_1 - b_1) \cup (E_2 - b_2));$$

take R to be the length-zero path when $b_1 = b_2$. Set

$$A_i = E_i - u_i b_i, \quad Z_i = D \cap (V(E_i) \setminus \{u_i, b_i\}).$$

Call E_i **light** if $|Z_i| \leq 1$ and **heavy** otherwise.

Adding the root edge in A_i gives a 2-connected graph containing E_i . An inner vertex of an end block has no neighbor in Q outside that block. Hence, for every $v \in V(E_i) \setminus \{b_i\}$,

$$d_{E_i}(v) = d_Q(v) = d_M(v).$$

Passing from E_i to A_i by deleting $u_i b_i$ affects only u_i, b_i . Thus (7.2) gives the degree hypotheses of Lemma 3.3 with $q = 3$. If $|Z_i| = 1$, its order condition also holds: $d_{E_i}(u_i) = d_M(u_i) \geq 3$ by (7.2), so $|V(A_i)| = |V(E_i)| \geq 4$. Thus each E_i supplies three admissible u_i, b_i -paths. A light family avoids c , whereas a heavy family may use c .

Suppose first that at least one end block is light. Then, for every choice of paths P_i from the two admissible families, at most one of P_1, P_2 contains c . Fix a y_1, y_2 -path L in T , orient R from b_1 to b_2 and L from y_1 to y_2 , and orient each P_i from u_i to b_i . Moreover,

$$V(P_i) \subseteq V(E_i) \cup \{c\}, \quad V(R) \subseteq V(Q) \setminus ((E_1 - b_1) \cup (E_2 - b_2)), \quad V(L) \subseteq V(T).$$

The interiors of E_1, E_2 are disjoint; R meets P_i only at b_i ; L is disjoint from $M \cup \{c\}$; and $y_1 \neq y_2$. If $b_1 = b_2$, that common cut vertex is the prescribed junction through the length-zero path R . Thus

$$P_1 + R + \overleftarrow{P_2} + u_2 y_2 + \overleftarrow{L} + y_1 u_1$$

is a simple cycle. If the selected path lengths are p_1, p_2 , its length is

$$p_1 + p_2 + |E(R)| + |E(L)| + 2.$$

The final three terms are fixed, so the two-variable form of Lemma 2.2 gives five admissible cycles, a contradiction.

It remains that both end blocks are heavy. Let K_i be obtained from E_i by adjoining c and all edges from c to $D \cap V(E_i)$. Since Z_i contains at least two vertices of $E_i - b_i$, c has at least two neighbors in E_i , and adjoining it through those distinct neighbors makes K_i 2-connected. Apply GHLM Theorem 3.1 with $q = 3$ to $K_i - cb_i$, rooted at the distinct vertices c, b_i . Adding the root edge preserves 2-connectivity, and deleting it affects only the roots. Every $v \in V(E_i) \setminus (D \cup \{b_i\})$ has degree at least four in E_i , while every $v \in (D \cap V(E_i)) \setminus \{b_i\}$ has degree at least three and gains its edge to c . Hence each side supplies three admissible c, b_i -paths.

For each i , the selected c, b_i -paths lie in $E_i \cup \{c\}$. Two such paths, one from each end block, have no common internal vertex: their $E_i - b_i$ portions lie in disjoint end-block interiors, and their only common vertex is c , together possibly with the common endpoint $b_1 = b_2$. The path R meets them only at b_1, b_2 . Consequently their union with R is a simple cycle.

The edge cb_i was deleted before applying the rooted theorem, so each selected path has length at least two. If their lengths are p_1, p_2 , the resulting cycle has length

$$p_1 + p_2 + |E(R)|.$$

The two-variable form of Lemma 2.2 again produces five admissible cycles, a contradiction. Hence every component of M is 2-connected. $\square$

7.2. Connectivity of the exterior

Two-link lemma. Let T be 2-connected, and let $X, Y \subseteq V(T)$ be disjoint two-element sets. Then T contains two vertex-disjoint (X, Y) -paths whose ends are distinct in X and distinct in Y , and whose internal vertices lie outside $X \cup Y$.

Proof. Add a new vertex a adjacent to the two vertices of X , and then a new vertex b adjacent to the two vertices of Y . Each new vertex is adjoined through two distinct neighbors, so the resulting graph remains 2-connected. By Menger's theorem, it contains two internally vertex-disjoint a, b -paths.

Delete a, b . The two remaining paths are vertex-disjoint. Their first vertices are distinct and therefore exhaust X , since a common first vertex would have been internal to both original paths. Similarly, their last vertices are distinct and exhaust Y . Neither remaining path can contain another vertex of $X \cup Y$ internally: that vertex is an endpoint of the other path, contradicting vertex-disjointness. These are the required links. $\square$

Lemma 7.2. The graph M is connected.

Proof. Suppose that Q_1, Q_2 are distinct components of M . Each has at least four neighbors in T , since deleting at most three such neighbors would separate that component from the other. By (7.1), no vertex of Q_i is incident with two attachment edges to T . Choose distinct $y_{1,1}, y_{1,2} \in N_T(Q_1)$ and then distinct

$$y_{2,1}, y_{2,2} \in N_T(Q_2) \setminus \{y_{1,1}, y_{1,2}\};$$

the latter choice is possible because $|N_T(Q_2)| \geq 4$. Choose corresponding attachment edges

$$u_{i,1}y_{i,1}, \quad u_{i,2}y_{i,2}$$

for $i = 1, 2$. By construction the four interface vertices $y_{i,j}$ are pairwise distinct. By (7.1), $u_{i,1} \neq u_{i,2}$ for each i , so both rooted endpoint pairs are distinct.

By Lemma 7.1, each Q_i is 2-connected. Set

$$A_i = Q_i - u_{i,1}u_{i,2}, \quad Z_i = D \cap (V(Q_i) \setminus \{u_{i,1}, u_{i,2}\}),$$

and call Q_i light if $|Z_i| \leq 1$ and heavy otherwise. Adding the root edge in A_i gives a 2-connected graph containing Q_i . Since Q_i is a component of M , every vertex retains all its M -neighbors, and deleting the root edge affects only the roots; thus (7.2) gives the degree hypotheses of Lemma 3.3 with $q = 3$. When $|Z_i| = 1$, its order condition also holds: $d_{Q_i}(u_{i,1}) = d_M(u_{i,1}) \geq 3$ by (7.2), so $|V(A_i)| = |V(Q_i)| \geq 4$. Hence each component supplies three admissible $u_{i,1}, u_{i,2}$ -paths; a light family avoids c , whereas a heavy family may use it.

Suppose first that at most one component is heavy. Apply the two-link lemma in T to

$$\{y_{1,1}, y_{1,2}\} \quad \text{and} \quad \{y_{2,1}, y_{2,2}\}.$$

After swapping the labels 1, 2 for Q_2 , and reversing its rooted path family if necessary, let the two vertex-disjoint links L_j join $y_{1,j}$ to $y_{2,j}$ for $j = 1, 2$. Orient a selected path P_i from $u_{i,1}$ to $u_{i,2}$. The links L_1, L_2 have disjoint vertex sets and have no internal vertex among the four interface vertices. A selected path P_i lies in $Q_i \cup \{c\}$. Since Q_1, Q_2 are distinct components and at most one of the two path families can use c , the paths P_1, P_2 are vertex-disjoint. They are also disjoint from both links, which lie in T .

It follows that

$$P_1 + u_{1,2}y_{1,2} + L_2 + y_{2,2}u_{2,2} + \overleftarrow{P_2} + u_{2,1}y_{2,1} + \overleftarrow{L_1} + y_{1,1}u_{1,1}$$

is a simple cycle. If the selected component-path lengths are p_1, p_2 , its length is

$$p_1 + p_2 + |E(L_1)| + |E(L_2)| + 4.$$

The connector contribution is fixed, so the two-variable form of Lemma 2.2 gives five admissible cycles.

Now suppose that both components are heavy. Choose one attachment edge $u_i y_i$ from each component with $y_1 \neq y_2$. Let K_i be obtained from Q_i by adjoining c and all edges from c to $D \cap V(Q_i)$. Since Z_i contains at least two nonroot vertices, c has at least two distinct neighbors in Q_i , so K_i is 2-connected. Since $c \notin V(J)$, the roots c, u_i are distinct, and $(K_i - cu_i) + cu_i$ contains K_i as a spanning subgraph and is 2-connected. Apply GHLM Theorem 3.1 with $q = 3$ to $K_i - cu_i$. Every nonroot outside D has degree at least four, and every nonroot in D gains its edge to c . Thus each side supplies three admissible c, u_i -paths.

A selected c, u_i -path lies in $Q_i \cup \{c\}$. Since Q_1, Q_2 are distinct components, the two selected paths meet exactly at their common endpoint c . Choose a y_1, y_2 -path L in T . It is disjoint from both exterior paths, and $y_1 \neq y_2$. Adjoining $u_1 y_1, L$, and $y_2 u_2$ therefore produces a simple cycle.

If the two selected path lengths are p_1, p_2 , its length is

$$p_1 + p_2 + |E(L)| + 2.$$

The two-variable form of Lemma 2.2 again gives five admissible cycles. Both cases are impossible; hence M is connected. $\square$

Since M is nonempty and connected, it has itself as its unique component; Lemma 7.1 therefore shows that M is 2-connected.

7.3. The final residue contradiction

The selected vertices $x, y \in T$ satisfy $d_T(x), d_T(y) \leq 3 < \delta(J)$. Since T is induced, each has a neighbor outside T . Choose $x', y' \in M$, respectively adjacent to x, y ; then $x' \neq y'$ by (7.1).

Define the exterior exceptional set and root-deleted graph by

$$Z = D \cap (V(M) \setminus \{x', y'\}), \quad A = M - x'y'.$$

By construction $x'y' \notin E(A)$. The graph $A + x'y'$ is 2-connected because it contains M as a spanning subgraph. Deleting the root edge affects only the root degrees, so (7.2) gives the hypotheses of Lemma 3.3 with $q = 3$. Moreover, $d_M(x') \geq 3$ by (7.2), so $|V(A)| = |V(M)| \geq 4$. We obtain three admissible x', y' -paths whose vertices lie in $V(M) \cup \{c\}$. Write their lengths as

$$a, a + d, a + 2d, \quad d \in \{1, 2\}.$$

Let $\mathcal{R} \subseteq \mathbb{Z}/5\mathbb{Z}$ be the three distinct residues of the selected x, y -paths in T . For a lifted x', y' -path P , adjoining xx' and $y'y$ produces an x, y -path whose internal vertices lie in $V(M) \cup \{c\}$. Since

$$V(T) \cap (V(M) \cup \{c\}) = \emptyset,$$

this exterior path meets every selected core x, y -path exactly at the distinct vertices x, y . Their union is therefore a simple cycle. If P is the i -th lifted path and the core path has residue $r \in \mathcal{R}$, its length is congruent to

$$a + id + 2 + r \pmod{5}, \quad 0 \leq i \leq 2.$$

Lemma 2.3, applied with a replaced by $a+2$, shows that one of these nine cycles has length divisible by five. This cycle lies in B , contradicting the standing assumption that B contains no such cycle. Hence the counterexample G_0 cannot exist.

8. Completion

The preceding contradiction proves the main result.

Theorem. Every nonempty finite simple graph of minimum degree at least five contains a simple cycle whose length is divisible by five.

Combining this theorem with the cases $k = 3$ and $k = 4$, proved by Chen and Saito^[2] and Dean, Lesniak, and Saito^[6], respectively, and the result of Luo, Ma, and Zhao for every $k \geq 6$ ^[8] gives the full conjecture:

Corollary (Dean's conjecture). For every integer $k \geq 3$, every nonempty finite simple graph of minimum degree at least k contains a simple cycle whose length is divisible by k .

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